

TaskWise: Enhancing Cognitive Performance in Task Management through Interactive Detail Enrichment

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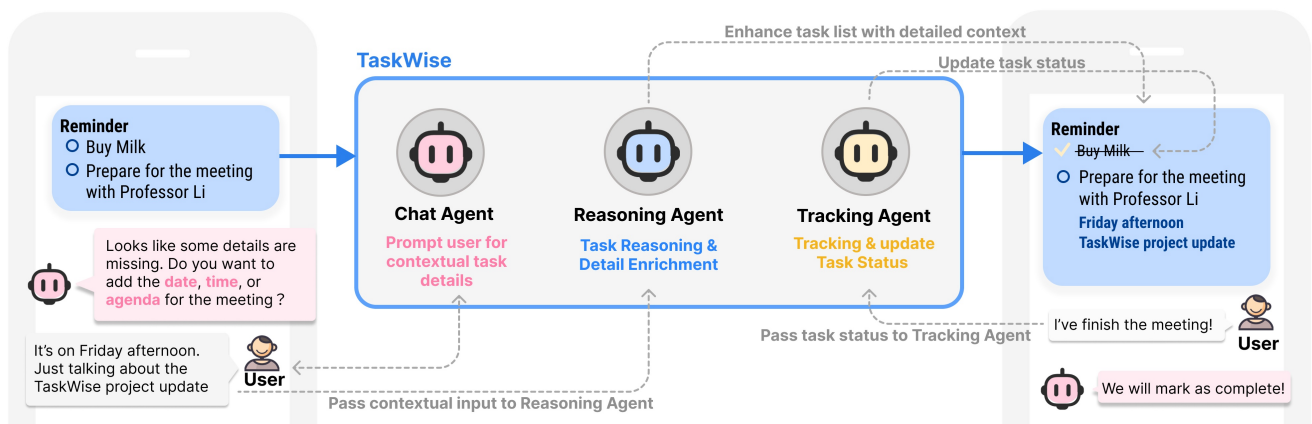


Figure 1: Schematic Diagram of the TaskWise System

Abstract

This study introduces TaskWise, an AI-driven to-do list management tool enhancing cognitive performance through interactive detail enrichment. TaskWise addresses the challenge of balancing simplified recording with sufficient initiation context by leveraging Large Language Models for real-time task analysis and proactive detail prompting, supporting effective task initiation and preventing key information omission. A seven-day, six-participant user study showed TaskWise's AI detail inference significantly improved users' task understanding, preparedness, and sense of control. AI-generated prompts served as an effective "cognitive scaffold," aiding memory recall and planning. Users also exhibited significant agency, strategically adopting AI-suggested details based on task context. These findings demonstrate AI's potential to assist task definition and enhance management experience, offering insights for

designing future intelligent systems that better support cognitive processes and empower user agency.

CCS Concepts

• **Human-centered computing** → **Empirical studies in collaborative and social computing**; *Empirical studies in HCI*.

Keywords

Task Management, Conversational Agent, Cognitive Enhancement

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1 Introduction

Enhancing human cognitive performance is a key focus in digital assistant design research [8, 12], as these assistants aid information acquisition, knowledge recognition, and decision-making, thereby improving functions like reasoning and memory [8]. To-do list management, a cognitively demanding real-world task, has driven

HCI exploration into innovative tool designs for better user performance. Early research on task management addressed interruptions and reattachment, exploring context-awareness [6], email integration [3], social reminders [11], emotionally-centered design [13], and personalized design [5] to understand how information presentation affects task management. Recently, focus has shifted to digital assistants that provide contextual or perceptual information, improving support for cognitive tasks like memorization and decision-making, and enhancing task initiation experiences [9, 14].

Recording and reviewing to-do lists are crucial for memory offloading and task reattachment, allowing users to connect tasks with relevant cues inspired during daily life [10]. However, users often over-simplify list entries for speed, leading to insufficient information and frustration at task initiation [3, 7]. This highlights a critical challenge: balancing simplified recording with the need for sufficient context for effective task execution. LLM-driven systems offering interactive detail enrichment present a promising solution. We developed TaskWise, an AI-enabled to-do list tool that uses LLMs to analyze task content and interactively help users enrich their records by prompting AI-generated cues and details. This aims to enhance their cognitive scaffolding, sense of preparedness, and control. We explored:

RQ1: How can AI-generated cue prompts improve users' cognitive performance (e.g., as cognitive scaffolds) and assist task management by influencing their task understanding and preparedness?

RQ2: What is the user experience of managing to-do lists with an AI system that interactively elicits and enriches task details, particularly concerning sense of control and contextual adaptation to AI suggestions?

Using a mixed-methods approach, six participants used TaskWise over seven days for task recording, status annotation, and retrieval via its conversational interface, choosing whether to accept AI-proposed cue prompts. Subsequent semi-structured interviews captured their experiences with the AI-driven detail enrichment and its impact. Participants reported a positive experience, noting that AI-driven interactive detail enrichment effectively supported their reflection and planning in early task formulation. This allowed them to gather more preparatory information, facilitating smoother task starts and enhancing their sense of task understanding and control.

2 TaskWise: System Design and Implementation

An initial survey of 30 participants revealed a common issue in to-do list management: inconsistent information processing standards between input and retrieval. To enhance recording speed for future tasks, users often simplify entries, omitting details like task duration, deadlines, or critical resources. Later, when initiating these tasks, users rely on these simplified records to recall the complete information. Over 80% of users reported using such simplification strategies and subsequently experiencing missing critical information at task initiation. This phenomenon inspired TaskWise, an AI-driven to-do list assistant, to balance recording convenience with sufficient contextual information for task initiation via an interactive detail enrichment mechanism, thereby improving users' sense of control. As shown in Figure 1, after a user submits a new

to-do item, TaskWise immediately records it to ensure the primary goal of quick capture is not interrupted. Subsequently, the system presents 2-3 suggested details as optional, non-intrusive prompts within the chat interface. Users can then choose to enrich their task with these suggestions, ignore them, or perform other list management actions via natural language.

Within TaskWise, the Chat Agent handles user inputs (to-do entries, updates, queries) and delivers detail prompts from item analysis. The Reasoning Agent analyzes tasks and generates relevant details. The Tracking Agent and database framework store to-do records and track progress. TaskWise features three key functionalities: (1) Item Analysis and Detail Generation: For new items, it analyzes and suggests related details to supplement user entries. (2) To-Do Item and Progress Updates: It helps users maintain to-do lists and track item progress. (3) Task Detail Addition: It allows users to add system-suggested details to specific to-do items. To enhance the relevance and accuracy of the suggestions, the Reasoning Agent's prompts are engineered to be grounded in the user's input. It primarily focuses on generating prompts that elicit common logistical details (e.g., asking for a time, place, or required documents), rather than generating speculative content itself. This approach mitigates the risk of providing incorrect information while still scaffolding the user's planning process.

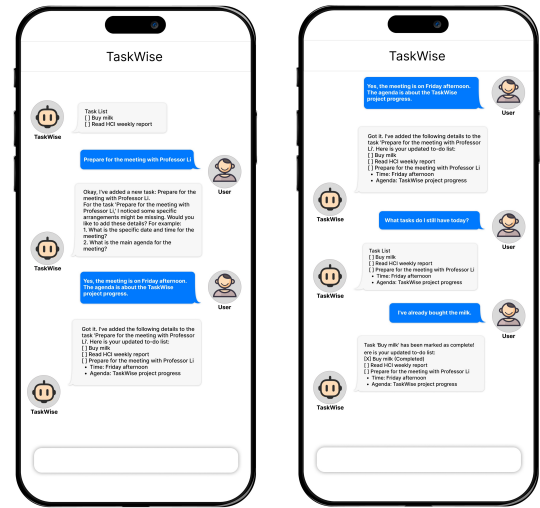


Figure 2: Schematic diagram of TaskWise user usage.

The system is built on Bytedance's Coze[2] multi-agent open platform, utilizing the CICI[1] tool invocation model, CICI[1] dialogue model, and data table plugins for core agent functionalities. TaskWise, a web application, is installable on PCs and mobile phones, enabling user interaction via voice or text through its conversational interface.

3 User Study

3.1 Participants

We disseminated participant recruitment posters for task management via social media platforms. Six participants, all with prior

experience using digital to-do list tools, voluntarily enrolled in our study. All participants utilized the TaskWise tool throughout a 7-day period and completed a post-interview. Each participant who completed the study received an incentive equivalent to \$14 USD as a token of appreciation for their participation. This research received ethical approval from our institution's review board. Participants were informed that their relevant data might be used for research analysis, and the confidentiality of their task management content and personal identity information was maintained throughout the study.

3.2 Procedure

Each study participant reported their frequency and preferences for daily work-related to-do list management by completing a survey. Prior to the commencement of the user study, researchers introduced TaskWise to the participants, explaining its functionalities to each individual. Through trial usage, all participants affirmed that the TaskWise prototype tool would not impede their normal work routines. Over the 7-day period in a real-world context, participants utilized TaskWise based on their actual needs, with the autonomy to either adopt or disregard the detail prompts suggested by TaskWise.

The data collection and analysis process referenced the emphasis on context inherent in the Experience Sampling Method[4]. Participants used TaskWise in their daily work and life, a format chosen for an authentic evaluation of the system's long-term effectiveness and user experience. Following the 7-day usage period, users participated in a semi-structured interview lasting approximately 30 minutes. Researchers interviewed participants regarding their experiences, any encountered issues, and their personal reflections on the task management process while using the tool. The researchers analyzed audio recordings, field notes, and responses from the semi-structured interviews. Employing Braun's thematic analysis method, at least two researchers conducted independent coding, subsequently collaborating to synthesize the user experience-related themes extracted from the interviews.

3.3 Results

3.3.1 Enhanced Task Understanding and Preparedness through AI-Generated Detail Prompts. AI-generated, project-specific detail reminders helped users initiate tasks more easily and prevent critical information omission, leading to greater clarity and reduced uncertainty. Of 80 recorded to-do items, users adopted AI-suggested details for 34, adding them to their task lists. TaskWise also effectively supported basic task recording and 65 progress updates. Participants reported that TaskWise's conversational recording process, with its explicit responses, provided a greater sense of "certainty." The Discussion section further explores how these AI-driven detail inference capabilities significantly enhance task management.

Interview feedback corroborated this enhanced experience. P1 noted, "It helps you organize things, like identifying the topic of a document you need to write," alleviating initial cognitive load. P2 stated, "After recording a task, it reminds me of things I might have forgotten," boosting confidence in task completeness. P6 shared, "I had a to-do item about an exam, and it prompted me to check the exam questions," improving focus on critical aspects. These

AI-driven prompts not only ensured comprehensive information recording but, crucially, enhanced users' task control. This led to a deeper understanding, better preparation, and greater confidence in task execution.

3.3.2 Increased Sense of Control and Confidence with AI-Assisted Task Enrichment. Through AI-assisted enrichment of task details, users experienced an enhanced sense of control over task management and increased confidence in the proper management and ultimate completion of tasks. Five participants explicitly stated their desire for their current to-do list tools to incorporate TaskWise's features of task-related detail association and prompting. This reflects an intrinsic user need for more comprehensive and controllable methods of task information management. The AI detail prompts helped users "prevent the omission of critical information," which directly impacted their experience of control. When users believe that crucial information is less likely to be lost, their anxiety regarding task management decreases, and their sense of control naturally strengthens. The act of users adopting AI-suggested details also reflects their perception that these suggestions contribute to improving the quality of task records, thereby boosting their confidence in the "executability" and "manageability" of the to-do item. AI assistance in detail recall and organization alleviated users' memory burden and planning stress. Consequently, users could focus more on the tasks themselves, experiencing a greater sense of control and managerial efficacy over their personal task lists.

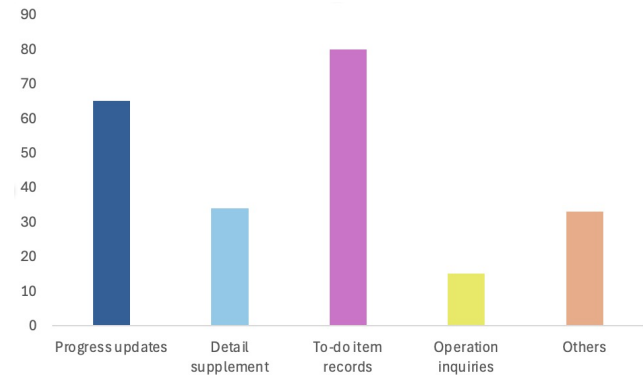


Figure 3: Statistics of user conversation intents.

3.3.3 Contextual Nuances in the Perceived Value of AI-Generated Details. While AI-generated detail prompts generally enhanced user task understanding, preparation, and control, the extent of this improvement and detail adoption propensity were moderated by task context (e.g., importance, frequency, familiarity). As shown in Figure 3, users adopted AI-provided details for 42.5% of the 80 recorded tasks. Participant feedback revealed this contextual moderation.

For periodic, fixed, or highly familiar tasks, users found concise records sufficient, perceiving less added value from AI prompts. P1 noted, "Some periodic tasks are fixed and only require brief records." P4 stated, "Some tasks are very familiar...so no detail prompts are needed, like financial reimbursement." Conversely, for tasks users deemed important..., AI prompts offered significantly more value. P3 expressed, "The tasks for which I adopted AI suggestions were

usually more important and needed more details." Users also tailored detail adoption to personal preferences for specific task categories, with P6 mentioning, "I want to manage tasks by category, and I prefer tasks related to physical exercise to have more details."

This demonstrates that users actively evaluated AI prompt relevance against their current task context, selectively adopting the most valuable details rather than passively accepting all suggestions. Such strategic interaction highlights user autonomy and control when leveraging AI tools for enhanced task management. The value of AI detail inference was thus most evident and appreciated in scenarios involving "important and meticulously planned" tasks.

4 Discussion

4.1 AI-Driven Detail Enrichment as Enhanced Cognitive Scaffolding in Task Management

The positive impact of AI prompts on task understanding and preparedness, as demonstrated in the "Results" section through user feedback (P1, P2, P6) and detail adoption behaviors, can be fundamentally understood as an effective "cognitive scaffold." By dynamically generating potential details relevant to the task, AI guides users towards deeper cognitive processing, facilitating associations from simple task descriptions to specific execution elements. This, in turn, aids memory retrieval, information organization, and preliminary planning, which are core functions of cognitive scaffolding.

Linking this scaffolding effect to Cognitive Load Theory, TaskWise reduces the user's cognitive load during the task initiation phase by offloading some of the mental work involved in information recall and organization. This allows users to allocate more cognitive resources to focusing on core execution. A brief comparison of TaskWise's interactive detail enrichment mechanism with other cognitive aids or prompting systems in HCI research reveals both similarities and distinctions. While many systems aim to support task management, TaskWise's potential advantages lie in its LLM-based dynamism and context-awareness. Unlike static checklists or predefined prompts, TaskWise can generate tailored and nuanced details that adapt to the specific content and context of the user's task. This dynamic, context-sensitive scaffolding has the potential to offer more personalized and effective cognitive support compared to more rigid, rule-based systems.

4.2 Emergence of User Agency and Trust in AI-Assisted Task Definition

Users' selective adoption of AI suggestions based on task context is a significant manifestation of their "agency." Far from passively accepting AI output, users act as active decision-makers and managers in the human-AI interaction, evaluating the value of AI suggestions and ultimately directing the final task definition. This agency is crucial for establishing healthy and effective human-AI collaborative relationships. User "trust" in the system and their experienced "sense of confirmation" can be more accurately attributed to the quality of AI suggestions—their accuracy, relevance, and inspirational nature—and the user's retention of ultimate control over these suggestions. Trust is established and deepened when AI consistently provides valuable assistance and users maintain control over its application. The enhanced "sense of control" experienced

by users is precisely the product of this agency combined with effective AI assistance.

4.3 Implications for Designing Adaptive and User-Centric AI Task Management Tools

Future AI task management tools, embodying Ubiquitous Computing principles, should integrate deeper into daily routines. Enhancing context-aware user agency is paramount: systems must leverage context-awareness for timely, relevant, and seamless intelligent assistance that empowers autonomous decision-making, all while ensuring user dominance. This demands deep personalization and dynamic situational adaptation beyond generic prompts. Tools need to understand individual work patterns and preferences, instantly adapting to nuanced contexts for personalized, dynamic support that integrates seamlessly into user rhythms. AI's role must evolve from a passive "auxiliary recorder" to a proactive "cognitive partner" that proactively senses needs, stimulates deep thinking, aids collaborative planning, and participates in co-defining and contextualizing task execution for ubiquitous cognitive enhancement. To avoid cognitive overload, designers should balance proactive assistance with user control. Offering customizable suggestion frequency and depth can prevent decision fatigue and ensure the tool remains a helpful cognitive aid, not a distraction.

5 Conclusion and Future Work

This study introduced TaskWise, an AI-driven tool that significantly enhances task management through interactive detail enrichment. Our findings demonstrate that TaskWise's AI-generated detail prompts improve users' task understanding, preparedness, and sense of control, acting as an effective cognitive scaffold. Users exhibited notable agency, strategically adopting these AI suggestions based on task context, highlighting a dynamic human-AI collaboration in task definition. These insights underscore the potential of AI to actively enrich cognitive processes in personal task management, offering valuable lessons for designing more intelligent and user-centric systems.

This study, while providing valuable qualitative insights, used a small sample and lacked a control group for comparative analysis. Future work will focus on enhancing AI adaptivity and personalization, evolving AI's role towards a proactive cognitive partner assisting with prioritization and planning, and empowering user agency in human-AI co-creation. Future work will involve larger-scale studies with control groups to rigorously assess the system's impact. Specifically, we aim to collect quantitative metrics such as task completion rates and times to measure potential productivity gains and determine statistical significance over baselines. This will allow for a more objective evaluation of TaskWise's impact on cognitive performance and overall user productivity, continuing to unlock AI's potential for task management tools that are deeply synergistic with human cognitive workflows.

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